

NATIVE DASHBOARD & VISUALIZATION FRAMEWORK

Development requirements for BI-style analytics inside the ticketing platform

A practical product, UX and engineering specification covering visualization components, interaction patterns, semantic data requirements, governance and phased delivery.

Recommended direction

Build a governed dashboard framework—not a collection of independent charts.

Prepared for: Development Teams

Document purpose

This document defines the minimum reusable visualization toolkit and supporting platform services required to create configurable, interactive dashboards within a ticketing platform. It is intended to guide product owners, UX designers, solution architects, front-end developers, data engineers, QA teams and implementation consultants.

Scope

- Dashboard builder and runtime viewer capabilities.
- Core visualization components and ticketing use cases.
- Data binding, calculations, filtering and interaction behavior.
- Security, performance, accessibility, export and governance requirements.
- MVP priorities, acceptance criteria and suggested architecture.

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1. Product vision and principles

The platform should allow authorized business users to assemble dashboards from governed data without requiring database access or code. Each dashboard must remain consistent with ticketing definitions, security rules and brand standards while supporting exploration from executive summary to transaction detail.

Principle	Requirement	Ticketing example
Governed self-service	Users build from certified dimensions and measures.	"Net Revenue" has one definition across POS, B2C, B2B and OTA.
Interactive by default	Selections filter or highlight related visuals.	Selecting "Waterpark" updates revenue, admissions and channel mix.
Progressive detail	Summary → drill-down → drill-through → transaction.	Venue revenue → product → performance → order.
Operationally useful	Near-real-time refresh, thresholds and action links.	Capacity utilization turns amber at 80% and red at 95%.
Responsive and accessible	Desktop, tablet and large control-room layouts; keyboard and screen-reader support.	Operations board remains legible on a wall display.

Key design decision

Separate the visualization engine from the semantic/query layer. Visuals describe how to render and interact; the semantic layer controls what fields, measures and permissions are available.

2. User roles and operating modes

Role	Primary need	Permissions
Executive / Viewer	Monitor KPIs and exceptions.	View, filter, drill, export allowed data.
Operations Manager	Monitor live attendance, queues, devices and capacity.	View live data, acknowledge alerts, navigate to operational actions.
Analyst	Explore trends and create dashboards.	Create/edit personal or shared dashboards from certified data.
Dashboard Designer	Publish governed templates.	Configure visuals, interactions, themes and page navigation.
Data Administrator	Manage datasets, measures and refresh.	Certify fields, define relationships, row security and cache policies.
Platform Administrator	Control tenancy and governance.	Manage permissions, audit, extensions and deployment lifecycle.

Required modes

- View mode: fast, safe consumption with filter, drill, tooltip and export controls.
- Edit mode: grid-based canvas, visual gallery, data fields, formatting and interaction panes.
- Presentation mode: full-screen, auto-refresh, simplified controls and large-format typography.
- Mobile layout mode: separately arrange the same governed visuals for narrow screens.

3. Core visualization library

The following components form the baseline visual library. “MVP” items should be delivered first because they cover most commercial and operational dashboards.

Visual	Purpose / minimum behavior	Ticketing examples	Priority
KPI / card	Single or multiple values; comparison, variance, trend sparkline, status icon.	Gross sales, net revenue, tickets sold, admissions today, refund rate.	MVP
Bar / column	Clustered, stacked, 100% stacked; sort, labels, reference line.	Sales by venue; tickets by channel; revenue by product.	MVP
Line / area	Continuous time axis, multiple series, zoom, missing-value behavior.	Hourly sales; daily admissions; membership renewal trend.	MVP
Combo	Column plus line; optional secondary axis with clear units.	Revenue columns with attendance line; sales with average ticket value.	MVP
Pie / donut	Part-to-whole with category limit and “Other” grouping.	Channel mix; payment tender share; ticket status mix.	MVP
Table	Exact detail, sorting, pagination/virtualization, totals, conditional format.	Orders, payments, refunds, admissions exceptions.	MVP
Matrix / pivot	Rows/columns, subtotals, hierarchy expand/collapse, aggregation.	Venue → product sales by month; channel by ticket type.	MVP
Slicer / filter control	List, search, dropdown, buttons, hierarchy and date range.	Venue, channel, product, performance date, customer segment.	MVP
Gauge / progress	Actual, target, bands and status; use selectively.	Capacity utilization; sales target achievement.	MVP
Funnel	Sequential stages and conversion rates.	Visits → cart → checkout → payment → issued ticket.	MVP
Waterfall	Positive/negative contributions to a total.	Gross sales – discounts – refunds – tax = net revenue.	MVP
Treemap	Hierarchical part-to-whole with drill.	Revenue by destination, venue and product family.	MVP
Scatter / bubble	Two numeric axes, size/color encoding, labels and zoom.	Price vs. conversion; attendance vs. spend per guest.	MVP

Visual	Purpose / minimum behavior	Ticketing examples	Priority
Map	Points/regions, clustering, legend and geographic privacy controls.	Sales by country/city; visitor origin; venue performance.	MVP
Heatmap	Color intensity across two dimensions.	Attendance by hour/day; seat or zone utilization.	MVP
Ribbon / rank	Rank changes across time.	Top products or venues by month.	MVP
Decomposition tree	Interactive breakdown of a measure by chosen dimensions.	Explain revenue decline by venue, channel, product and discount.	MVP
Narrative / insight text	Data-bound text, comparisons and generated highlights with traceability.	"Admissions are 12% above last Tuesday."	MVP

4. Dashboard interaction framework

A Power BI-like experience depends on consistent interactions shared by every visual. These capabilities should be implemented centrally, not separately within each chart.

Capability	Required behavior	Acceptance example
Cross-filtering	Selecting a mark filters subscribed visuals using the selected dimension values.	Clicking B2C filters KPIs and order table to B2C.
Cross-highlighting	Preserve totals while emphasizing the selected contribution.	Selecting a venue highlights its contribution in monthly sales.
Drill-down / up	Navigate a declared hierarchy without leaving the visual.	Region → venue → product → performance.
Drill-through	Open a target page with filter context and a visible back action.	From refund KPI to refund transaction detail.
Tooltips	Default values plus configurable rich tooltip content; keyboard accessible.	Hover/tap a date to see sales, tickets, ATP and variance.
Visual/page/report filters	Clear scope and visible active-filter state.	Performance date at page level; tenant at security level.
Bookmarks and reset	Save view state and restore the designed default.	Operations preset: today, all venues, live mode.
Zoom, brush and select	Time-range selection and multi-select with predictable modifiers.	Drag across 10:00–14:00 to inspect peak operations.
Navigation and actions	Buttons for pages, URLs and authorized platform workflows.	Open order, device incident or capacity configuration.
Refresh state	Show data-as-of time, loading, stale, partial and error states.	"Updated 15 sec ago"; stale warning after threshold.

Interaction safeguards

- Display filter breadcrumbs and provide Clear / Reset controls.
- Prevent ambiguous double axes, misleading truncation and silent filter propagation.
- Selections must be encoded in the URL or saved state where sharing is permitted.
- Every visual needs empty, loading, error, no-permission and data-stale states.

5. Dashboard builder requirements

Builder area	Minimum functions
Canvas	12-column responsive grid; drag, resize, align, distribute, layer, lock, duplicate, group; undo/redo.
Visual gallery	Searchable categories, preview, supported data-shape guidance and recommended visual.
Data pane	Datasets, certified fields, measures, data types, hierarchies, search, favorites and descriptions.
Field wells	Role-based drop zones such as Axis, Legend, Values, Small Multiples, Tooltips and Filters; validation messages.
Formatting pane	Title, subtitle, axes, labels, legend, colors, background, border, spacing, number/date format and responsive rules.
Analytics pane	Constant/average/min/max/median/percentile lines, trend line, target, forecast and anomaly settings where supported.
Filter pane	Visual, page and report scopes; basic/advanced/relative date/Top N filters; locked or hidden filters.
Interaction editor	Designer chooses filter, highlight or no interaction between visual pairs.
Theme manager	Tenant theme tokens, accessible palettes, fonts, status colors; JSON import/export or equivalent schema.
Pages and navigation	Add, rename, duplicate, reorder, hide; page size; navigation buttons and bookmarks.
Publish lifecycle	Draft, preview, validate, publish, version, rollback, clone and promote between environments.

Usability rule

The builder should recommend a visual based on the selected fields, but the user retains control. Invalid combinations must explain what is missing—for example, “A line chart requires a date/category axis and at least one numeric measure.”

6. Semantic model and calculations

Dashboards must query a business-friendly semantic model rather than raw transactional tables. The model should define dimensions, measures, relationships, hierarchies, formats and security once for reuse.

Capability	Requirement
Certified datasets	Named subject areas such as Sales, Orders, Admissions, Capacity, Customers, Membership, F&B and Devices.
Dimensions	Date/time, tenant, brand, region, venue, department, workstation, channel, product, ticket type, performance, customer and promotion.
Measures	Reusable aggregations with business definitions, owner, format, lineage and test cases.
Calculation types	Sum, count, distinct count, min/max, average, ratio, variance, percent of total, running total, moving average, rank and period comparison.
Time intelligence	Today, week/month/quarter/year, previous period, previous year, rolling windows and fiscal calendar.
Hierarchies	Declared drill paths; e.g., Tenant → Region → Venue → Department → Workstation.
Query contract	Visual sends dataset, dimensions, measures, filters, sort, limit and calculation request; service returns typed tabular data plus metadata.
Data quality	Freshness, completeness and reconciliation indicators available as measures and visual status.

Metric definition examples

Metric	Illustrative definition	Notes
Gross Sales	Sum of issued sales before discounts/refunds, using configured tax treatment.	Definition must state whether tax and fees are included.
Net Revenue	Gross Sales – Discounts – Refunds, adjusted for cancellations/voids per finance policy.	Keep operational sales and recognized revenue distinct.
Average Ticket Value	Ticket revenue ÷ eligible tickets sold.	Exclude complimentary tickets if policy requires.
Admission Conversion	Admitted tickets ÷ valid issued tickets for the selected visit period.	Use visit date, not only sale date.
Capacity Utilization	Consumed/reserved capacity ÷ sellable capacity.	Define treatment of holds and blocked inventory.
Channel Conversion	Completed paid orders ÷ checkout sessions.	Requires web/session telemetry.

7. Ticketing dashboards and required KPIs

Dashboard	Core KPIs	Recommended visuals
Executive Performance	Gross/net revenue, tickets, admissions, ATV, refund %, target variance.	KPI cards, line/combo, bar, waterfall, matrix.
Sales & Channel	Sales by POS/B2C/B2B/OTA, conversion, basket size, discount use.	Stacked bars, funnel, trend, table, slicers.
Admissions & Capacity	Entries, no-shows, utilization, available-to-sell, peak hour.	Cards, line/area, heatmap, gauge, detail table.
Product & Pricing	Revenue/volume/margin by product, price point and promotion.	Bar, treemap, scatter, waterfall.
Customer & Membership	New/renewed/expired members, visits, retention, customer origin.	Cards, cohort/trend, map, matrix.
Finance & Reconciliation	Payments, refunds, chargebacks, tender variance, unsettled transactions.	Waterfall, cards, matrix, exception table.
Operations Control Room	Live admissions, queues, device health, incidents and capacity alerts.	Live cards, time series, status grid, map/venue layout.
B2B / OTA Partner	Partner sales, commission, credit utilization, API failures.	Cards, bar, trend, matrix, exception table.
F&B / Retail	Revenue, transactions, items, attachment rate, stock/service exceptions.	Cards, bar, hourly heatmap, table.

8. Suggested architecture and non-functional requirements

Layered architecture

1. Definition Layer — Dashboard Definition Store Versioned JSON metadata describing pages, visuals, layout, data bindings, filters, theme, and cross-visual interactions. Supports diff/merge and rollback so dashboard changes can go through review before publishing.
2. Modeling Layer — Semantic Metadata Service Owns datasets, dimensions, measures, hierarchies, display formats, lineage, and certification status. This is the single source of truth for "what a field means" across all dashboards.
3. Access Layer — Query Service Validates incoming requests, enforces tenant, row-, and column-level security, translates dashboard queries into executable queries, applies row/result limits, and

returns typed results. This is the only place security is enforced — no other layer should be able to bypass it.

4. Logic Layer — Calculation Engine Executes governed (centrally defined) measures plus safe, sandboxed visual-level calculations. Tightly coupled to the Query Service's query planning, since calculation and query execution are rarely cleanly separable in practice.
5. Presentation Layer — Visualization Runtime A reusable visual component SDK, a shared interaction bus (cross-filtering, drill-through), accessibility support, and export hooks for individual visuals.
6. Performance Layer — Cache & Refresh Coordinator Handles result caching, cache invalidation, scheduled refresh, live/polling updates, and freshness metadata. Cache keys must include the security context (tenant/row/column permissions), not just the query shape, to prevent cross-tenant data leakage through cached results.
7. Output Layer — Export & Rendering Service Produces CSV/XLSX detail exports, image/PDF snapshots, and scheduled delivery — all gated by the same authorization rules enforced in the Query Service.
8. Governance Layer — Audit & Telemetry Captures usage, performance, publication, export, and security events across every layer above, for compliance and operational monitoring.

Two additions worth including

9. Ingestion / Connector Layer Sits before the Semantic Metadata Service — handles how raw data actually lands (connectors, pipelines, transformation) before it's modeled. Not implicit in the original list, but every real BI platform needs it as its own layer.
10. Authoring & Publishing Workflow Governs how a dashboard moves from draft → review → certified → published, tying together the Definition Store, Semantic Metadata Service (certification status), and Audit layer. Prevents unreviewed or uncertified content from reaching production dashboards.

Key design rules to call out explicitly in the doc

Security enforcement lives only in the Query Service; the Calculation Engine, Cache Coordinator, and Export Service must all route through it rather than re-implementing checks.

Cache invalidation and cache keys must be security-context-aware.

The Definition Store and Semantic Metadata Service should be versioned independently — dashboard layout changes and data model changes are different change types with different review needs.

Non-functional targets

Area	Recommended baseline
Performance	Dashboard shell ≤ 2 s; cached visuals ≤ 2 s; uncached common queries ≤ 5 s at p95; progressive rendering.
Scale	Virtualized tables; server-side aggregation; configurable row/cardinality limits; query cancellation and concurrency controls.
Freshness	Dataset-specific SLA; visible “as of” timestamp; polling/WebSocket only where operational need justifies it.

Area	Recommended baseline
Reliability	Graceful partial failure; retry; circuit breaker; one failed visual must not blank the page.
Observability	Query ID, dataset, duration, rows, cache status, errors and client render time.
Browser/device	Current major desktop browsers; tablet responsive layout; large-screen presentation mode.
Extensibility	Versioned visual SDK and manifest; sandbox custom code; capability and permission declaration.
Localization	Locale-aware date, time, number and currency; RTL-ready; venue timezone handling; 2/3 decimal currencies.

9. Governance, security and accessibility

Security and governance

- Tenant isolation at every query and cache key; no client-supplied tenant trust.
- Role-based dashboard permissions plus row-level and column-level data security.
- Sensitive-field classification, masking and export restrictions for personal/customer data.
- Audit create, edit, publish, share, export and administrative actions.
- Certified dataset/measure badges, owner, glossary, lineage and deprecation workflow.
- Custom visuals must be allow-listed, signed/versioned, sandboxed and prevented from arbitrary network/data exfiltration.

Accessibility and visual integrity

- Keyboard navigation, visible focus, screen-reader names and logical tab order.
- Do not rely on color alone; use labels, shapes/icons and accessible status text.
- WCAG-aligned contrast, scalable text and accessible data-table alternative.
- Alternative text for visuals and export; motion reduction where animation exists.
- Zero-baseline defaults for bars unless an explicit analytical reason is shown; clear units and axis scales.

10. Acceptance criteria

MVP acceptance criteria

- A designer can create, save, preview, publish, clone and version a multi-page dashboard without code.
- At least eight core visual types bind to certified dimensions/measures and handle loading, empty, error and stale states.
- Page/report/visual filters, date slicers, cross-filtering, drill-down and drill-through behave consistently.

- All queries enforce tenant and row-level security; exports cannot bypass visible permissions.
- Dashboards support tenant themes, conditional formatting, accessible palettes and keyboard navigation.
- Performance targets are measured with realistic high-volume ticketing data.
- Definitions and totals reconcile against approved sales, payment and admissions reports.

Appendix A — Visual selection guide

Question	Preferred visual	Avoid / caution
What is the current value?	Card or KPI	Do not use a gauge unless target/range matters.
How do categories compare?	Bar/column	Avoid pie with many categories.
How is it changing over time?	Line; area for magnitude	Avoid categorical spacing on continuous dates.
What contributes to the total?	Stacked bar, donut (few items), treemap	Avoid 3D effects.
What caused the change?	Waterfall or decomposition tree	Do not imply causation from correlation.
How are two measures related?	Scatter/bubble	Show sample size and outliers.
Where did it happen?	Map	Use only when location adds meaning; protect privacy.
What are the exact records?	Table/matrix	Use virtualization and server-side totals.
Where are peaks across two dimensions?	Heatmap	Provide legend and accessible values.

Appendix B — Dashboard definition (illustrative)

Store dashboard configuration as versioned metadata. The exact schema may differ, but it should express identity, data binding, layout, formatting, interactions and governance without embedding executable code.

```
dashboard: { id, version, datasetId, pages[], themeId, defaultFilters[], permissions }
```

```
page: { id, title, layout, visuals[], pageFilters[], bookmarks[] }
```

```
visual: { id, type, position, bindings, calculations, sort, filters, format, interactions, accessibility }
```

```
binding: { role, fieldId | measureId, aggregation, format }
```

Appendix C — Key QA scenarios

- Filter propagation across mixed charts and tables, including reset and bookmark restore.
- Hierarchy drill with subtotals, null/unknown members and security-trimmed nodes.
- Timezone boundaries, daylight-saving behavior, visit date versus transaction date.
- Currencies with two and three decimal places; locale formatting and RTL layout.
- Refunds, voids, cancellations, complimentary tickets, partial payments and negative values.
- Large datasets, high-cardinality dimensions, slow query cancellation and partial failure.
- Permission changes after a dashboard is saved; prohibited field and export behavior.
- Accessibility with keyboard only, screen reader, high zoom and non-color status cues.